OLIVER LARSEN

Software Engineer - AI

Willing to relocate to London, stop my master's studies, instructing, research project, and Carboplanner startup immediately

Oliver1703dk@hotmail.com | +45 60483836 | Odense C, Denmark
oliver-larsen-b7bb89243 | Oliver1703dk | @oliverlars38054

PROFILE

I want to work on the most truth-seeking AI.
I'm a software engineer who seeks out difficult problems and environments where focus, discipline, and curiosity matter. I push myself both mentally and physically, choosing challenges that force growth and deeper thinking. I approach my work with humility, a truth-seeking mindset, and a willingness to question my own assumptions. Clear communication is important to me, I communicate deliberately and effectively, while working to improve every day, as I also have a stutter. I value learning from others and contributing in a hands-on, high-ownership way. I thrive where complexity is high, stakes are real, and the goal is to build something exceptional.

EXPERIENCE

Student Software Research Assistant
University of Southern Denmark
10/2024 – present | Odense (hybrid)

- Flood-detection research: Camera + sensor fusion on distributed edge devices (Pi + Jetson) with runtime adaptation manager for ML inference.

Teaching Assistant
University of Southern Denmark
09/2023 – present | Odense

- Teaching in Object Oriented Programming, Data Management, and Computer Systems. Delivered lectures for 300+ university students.

Co-founder & Software Engineer
CarboPlanner
10/2024 – present | Odense (hybrid)

- Co-led full-stack development for mobile, web, and backend; Focused on backend development including nutrition recommendation algorithms and agentic ai implementation; participated in raising 1,090,000 DKK.

Soldier
The Royal Life Guards
08/2021 – 03/2022 | Copenhagen

- Completed 8 months of military service: 4 months soldier conscription and 4 months of guard for the royal danish family.

EDUCATION

Master's in Software Engineering
University of Southern Denmark
08/2025 – ongoing | Odense

Bachelor's in Software Engineering
University of Southern Denmark
09/2022 – 06/2025 | Odense

- Software Engineer (BSc Eng). Grade average: ∼9.5 (A).
- Thesis In LLMs as Recommender Systems; Grade: 12 (A+).

High School Diploma
Slagelse Gymnasium
08/2018 – 06/2021 | Slagelse

- Specialized in Math, Physics, Chemistry; Grade average: ∼9 (A-).
- Thesis in Math and Informatics about Enigma cryptography; Grade: 12 (A+).

ACHIEVEMENTS

Ironman Copenhagen 140.6
Triathlon
08/2025

- Completed 3.8 km swim, 180 km bike, 42.2 km run.

Coach Training, Stammer Program
International Program
2020

- Coach in international mentoring program supporting people with speech disabilities.

SKILLS

Fine-Tuning (LoRA/PEFT)
In-Context Learning & Prompt Engineering
Multi-Agent Systems
Distributed Edge ML Architecture
Machine Learning
LLM Integration
Full-Stack Development
Software & Database Architecture

PROJECTS

CarboPlanner Nutrional Guidance System

Startup Project

2024 – present

CarboPlanner is a sports technology startup developing digital nutrition solutions tailored for football clubs. I co-led the technical implementation across a React-based Coach web module, a Flutter mobile app suite (iOS/Android) and a NestJS backend with MongoDB, enabling individualized meal plans, macronutrient recommendations, role-based access, and management for athletes and staff. I designed the end-to-end recommendation pipeline to deliver precise, adaptive nutrition guidance. Beyond engineering, I contributed to securing 1,090,000 DKK in funding and are supporting current live pilot testing with elite football clubs worldwide.

Hybrid LLM-Based Meal Recommender System

Bachelor's Thesis

2025

Built a hybrid meal recommender utilizing LLM, for CarboPlanner by integrating in-context learning with LoRA/PEFT fine-tuning of Mistral-7B-Instruct. Implemented as a Flask microservice connected to the CarboPlanner Flutter app, NestJS backend, and MongoDB, with in context-learning and PEFT dual feedback loops for short- and long-term personalization adaptation. Achieved 46.7% accuracy and 100% format adherence on 1,000 test prompts, outperforming accuracy baselines of 41.5% without fine-tuning, and 20% random baseline.

Flood Detection System

Research Project

2024 – present

I developed a real-time flood detection architecture for off-road vehicles on distributed edge devices. A Raspberry Pi capture imagery and sensor data (temperature, humidity, pressure), while classification tasks (YOLOv8 inference and sensor processing) are routed via MQTT to either a Pi or a Jetson AGX Orin. A finite state machine runtime adaptation manager switches the sizes of the YOLO models and forwards processing tasks to Pi or Jetson to balance accuracy, latency, and energy. Multimodal fusion combines image predictions with sensor anomalies, using adaptive baselines for drift handling.

Stock Analysis System

Personal Project

2025

I built an online stock analysis hub applying logic from multi-agent LLM trading research, extended with my own implementation for improved execution. The system ingests market data, runs LLM analyses, and outputs explainable reports and *buy*, *sell*, and *hold* targets with risk-aware *stop-loss* levels. A Next.js frontend displays analysis cards, while a Flask API with MongoDB manages pipelines and persistence. To operate within free hosting limits, Celery and Upstash Redis queue asynchronous jobs for efficient use of limited compute.

Spark Romance Relationship App

Startup Project

2023 – 2024

I built Spark Romance a relationship-enhancing mobile application designed to help couples strengthen their connection by encouraging thoughtful actions and shared activities. The system was developed using a microservice-inspired architecture with four Flask-based APIs and a Flutter frontend, backed by a complex PostgreSQL database. This design emphasized scalability and modularity, making it straightforward to extend features and handle growth.

Enigma Simulator

High School SRP

2021

As my final project in high school, I developed a fully working Enigma machine simulator using the drag-and-drop programming language Scratch. The project was deliberately over-engineered compared to the simplicity of the language, replicating rotor wiring, stepping, and plugboard logic. It demonstrated an early ability to tackle complex system design and problem-solving with limited tools.